

REVIEW ARTICLE OPEN ACCESS

Smart Waste, Smarter World: Exploring Waste Types, Trends, and Tech-Driven Valorization Through Artificial Intelligence, Internet of Things, and Blockchain

Segun E. Ibitoye^{1,2,3}  | Amit K. Ball⁴ | Raul V. Rodriguez⁴ | Olalekan A. Olayemi⁵ | Pepluis Esteve¹ | Peter O. Omoniyi⁶ | Isaac K. Adegun² | Rasheedat M. Mahamood^{3,7} | Esther T. Akinlabi⁸

¹School of Technology, Woxsen University, Hyderabad, India | ²Department of Mechanical Engineering, Faculty of Engineering and Technology, University of Ilorin, Ilorin, Nigeria | ³Department of Mechanical Engineering Science, Faculty of Engineering and the Built Environment, University of Johannesburg, Auckland Park, South Africa | ⁴School of Business, Woxsen University, Hyderabad, India | ⁵Department of Aeronautics and Astronautics, Faculty of Engineering and Technology, Kwara State University, Malete, Kwara, Nigeria | ⁶Department of Mechanical Engineering, Howard University, Washington, DC, USA | ⁷Department of Mechanical and Construction Engineering, Faculty of Engineering and Environment, Northumbria University, Newcastle, UK | ⁸Department of Mechanical Engineering, Colorado State University, Fort Collins, Colorado, USA

Correspondence: Segun E. Ibitoye (ibitoye.s@unilorin.edu.ng) | Amit K. Ball (amitball.dgp@gmail.com)

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ABSTRACT

Global municipal solid waste generation is projected to exceed 3.8 billion tonnes annually by 2050. This makes the need for smart, inclusive, and scalable waste valorization systems more urgent than ever. This review critically explores the shift from conventional waste management to intelligent, technology-driven solutions aligned with circular economy goals. Key findings highlight the transformative role of digital tools in waste classification, forecasting, and real-time monitoring. Long Short-Term Memory models achieved up to 94% accuracy in biogas prediction, while XGBoost demonstrated 98.5% accuracy in solid waste generation forecasting. Deep learning systems have reached classification accuracies of 83.11% across 28 recyclable categories and mean average precision scores up to 63% in complex waste detection. Despite promising advances, challenges such as data quality, regulatory hurdles, and system interoperability persist. This article contributes both a conceptual and practical blueprint for stakeholders, positioning smart waste valorization as a strategic opportunity to drive innovation.

1 | Introduction

Rapid industrialization, technological advances, and population growth are intensifying global environmental challenges. This is most visible through the escalating volume and complexity of waste generated across economic sectors (Pedro et al. 2024;

Ullah et al. 2025). Traditional disposal practices are now increasingly unsustainable, once suited to predominantly organic waste streams. These include dumping (Sridhar et al. 2024), open burning (Bamigbade et al. 2025), incineration (Ling Wen Xia et al. 2024), and burying. Modern waste is heterogeneous, encompassing plastics (Olazar et al. 2024), e-waste (Kwon

Abbreviations: AD, anaerobic digestion; AI, artificial intelligence; CE, circular economy; CNN, convolutional neural network; DL, deep learning; DLT, distributed ledger technology; EEE, electrical and electronic equipment; GIS, geographic information system; IoT, internet of things; JSON-LD, JavaScript Object Notation for Linked Data; LSP, large-scale pilot; mAP50, mean average precision at 50%; ML, machine learning; MRF, material recovery facility; MSW, municipal solid waste; SDGs, sustainable development goals; WMR-DL, waste management recovery-deep learning framework; WV, waste valorization.

Segun E. Ibitoye and Amit K. Ball contributed equally to this work.

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et al. 2024), construction leftovers (El et al. 2024), industrial residues (Nadaleti et al. 2024), and hazardous materials (Kwon et al. 2024).

Global waste generation is rising rapidly. However, estimates vary depending on the scope considered. For municipal solid waste (MSW) alone, the UNEP Global Waste Management (WM) Outlook estimated growth from approximately 2.1 BT to about 3.8 BT by 2050 if current practices persist (UNEP 2024). This trajectory underscores the urgency of adopting sustainable, state-of-the-art WM solutions. When broader categories of waste, such as industrial, agricultural, mining, and construction and demolition residues, are included, substantially higher totals are reported, estimating around 20 BT and projecting up to 46 BT by 2050 under current trends (Maalouf and Mavropoulos 2023).

Waste valorization (WV) is the transformation of unwanted materials into value-added products, offering one of the most viable pathways for reducing environmental burden while supporting circular economy (CE) goals (Ariyanti et al. 2025; Bint-e-Zahra 2024). By reconceptualizing waste as a resource, WV reduces reliance on virgin raw materials, mitigates greenhouse gas emissions, and supports socioeconomic development (Elmeknassi et al. 2024; Preethi et al. 2024). Its applications include biodiesel production (Ling Wen Xia et al. 2024), biochar generation (Huang et al. 2025), biogas production (Nesterov et al. 2024), and resource recovery from e-waste. Resource recovery from e-waste enhances global energy security. In the agricultural sector, the transformation of biomass feedstock into biochar and compost improves soil fertility and crop production, promoting sustainable agriculture and food security (Bano et al. 2025; Javed et al. 2025).

Numerous thermal, chemical, and biological waste conversion technologies exist. These include pyrolysis, gasification, and anaerobic digestion (AD) (Buentello-Montoya et al. 2025; Mazzanti et al. 2025; Shen and Wu 2025; Zhang et al. 2025). However, this review focuses on digital innovations. These innovations are increasingly shaping how WV is designed, managed, and scaled. The integration of three key technologies is revolutionizing waste management: the Internet of Things (IoT), artificial intelligence (AI), including machine learning (ML) and deep learning (DL), and blockchain technologies. These technologies are transforming waste identification, classification, tracking, and governance (Ansari et al. 2024; Haque et al. 2024). These digital tools enhance operational efficiency, transparency, and trust across the WM chain, enabling data-driven decision-making and more resilient CE models.

The deployment of digital technologies in WV aligns directly with global policy frameworks. For instance, the European Union's CE Action Plan emphasizes digitalization as a lever for reducing resource loss (EU 2025). Also, the United Nations Sustainable Development Goals (SDGs) call for substantial reductions in waste generation through prevention, recycling, and reuse (UN 2024). Smart-bin and routing pilots reduce MSW and improve resource efficiency, supporting SDG 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production). Blockchain-based traceability and credit systems further strengthen SDG 12 while reinforcing SDG 16 (Peace, Justice and Strong Institutions) through greater

transparency and accountability. IoT-monitored pyrolysis and biochar production contribute to SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action) via the generation of renewable fuels and lowering emissions.

WV faces several challenges despite its numerous benefits and widespread support. These include market demand, regulatory barriers, economic viability, technology limitations, and societal acceptance (Karim et al. 2025; Luo et al. 2024; Ullah et al. 2025). Collaboration amongst several sectors, including government, industry, higher education, and civil society, is necessary to address these issues. This article offers a thorough overview of WV by examining the different types of waste. Further, it assesses the extent of valorization processes and the opportunities and problems posed by contemporary global WM practices and emerging technology. Additionally, this review synthesizes current knowledge on how AI/ML/DL, IoT, and blockchain are applied to WM, focusing on three priority waste streams: MSW, e-waste, and biomass residues. It highlights real-world applications, challenges, and gaps, and proposes a unified architecture for scaling digital WV solutions. The article is structured into seven sections. The review methodology is presented in Section 2. Section 3 presents an overview of global WM challenges, while Section 4 discusses innovative approaches to WM. The prospects, recommendations, and conclusions are presented in Sections 5–7, respectively.

2 | Methodology

A multi-phase technique was employed to guarantee a thorough, methodical, and rigorous approach in conducting this review. This covers thorough search procedures, exact exclusion and inclusion criteria, and data extraction and analysis. The technique was created to identify, evaluate, and compile the most recent and pertinent research on WM. Also, it evaluates the implementation of cutting-edge technologies, specifically blockchain, AI, and IoT in the WV.

2.1 | Literature Search Strategy

The literature search was carried out to identify studies pertinent to the scope of smart WM and technology-driven WV using a systematic approach. The search focused on policy documents, academic manuscripts, and peer-reviewed journals from 2020 to 2025. The IEEE Xplore, Web of Science, Scopus, and ScienceDirect databases were selected as the primary sources of academic and peer-reviewed articles for the review. Additionally, Google Search was utilized to identify high-relevance articles, policy documents, and case studies not available on the ScienceDirect, Web of Science, Scopus, and IEEE Xplore databases. The search strategies employed broad sets of keywords to maximize search coverage. These included terms related to waste management, waste valorization, CE, and digital technologies (AI, ML, DL, IoT, blockchain). The general query was: (“waste management” OR “waste valorization” OR “circular economy”) AND (“artificial intelligence” OR “machine learning” OR “deep learning” OR “IoT” OR “Internet of Things” OR “blockchain”) AND (2020–2025). This string was adapted as necessary to each database's syntax. A wide range

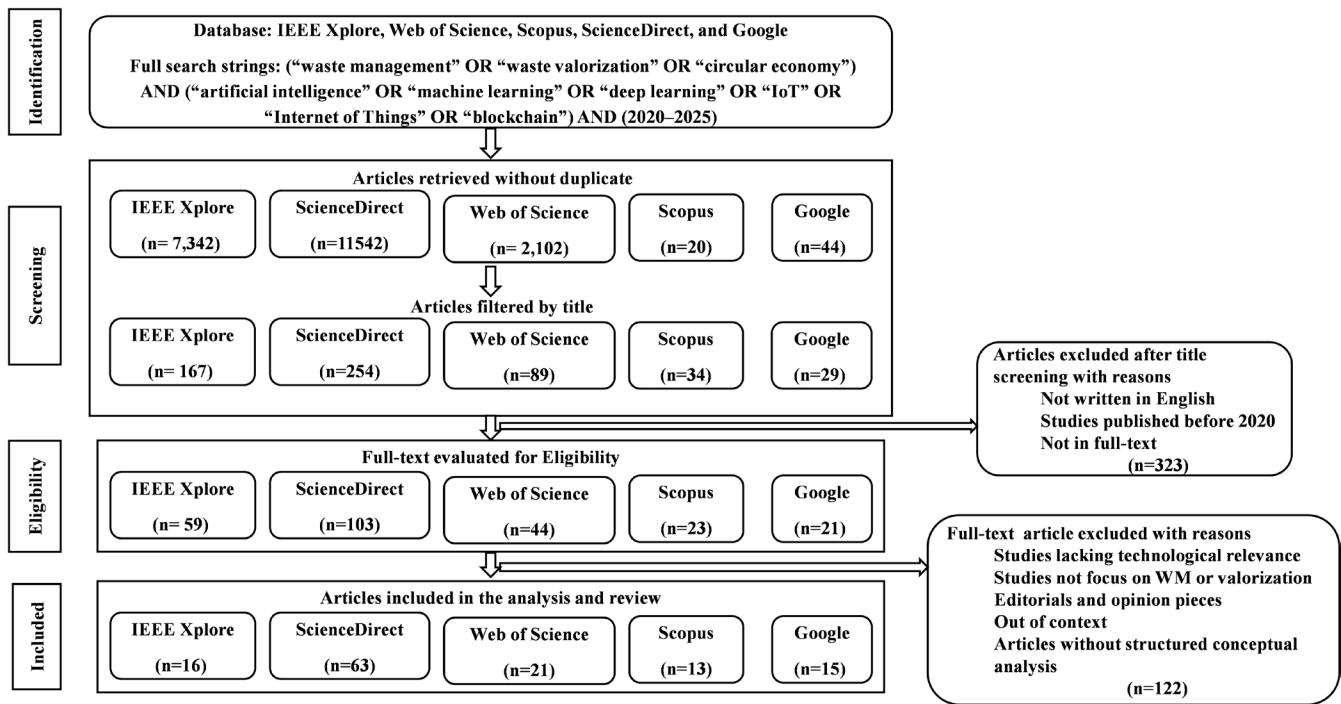


FIGURE 1 | The PRISMA framework of the review process.

of relevant literature was captured by using combinations of these terms encompassing technical, environmental, energy, economic, and policy perspectives.

2.2 | Inclusion and Exclusion Criteria

Sets of inclusion and exclusion criteria were applied following the guidelines reported by Balali et al. (2023) and Vlachokostas et al. (2021). The inclusion criteria focused on studies that were published in peer-reviewed journals and conference proceedings, studies that explored recent advancements in AI, IoT, ML, DL, and Blockchain as applied to waste types, WM trends, or valorization pathways. Only articles written in English and available in full text were considered. Further, technical reports and policy documents directly related to at least one of the three core themes: waste type classification, technology integration, and valorization impact were considered. Conversely, studies were excluded if they lacked technological relevance, did not focus on WM or valorization, and were published before 2020 without being foundational. Nevertheless, certain articles published before 2020 that were directly pertinent to the subject were retained in the review. Editorials, opinion pieces, and papers without empirical or structured conceptual analysis were also excluded.

2.3 | Data Extraction and Thematic Analysis

The data extraction process was conducted in a structured and manual manner following the identification of pertinent literature. Each selected study was reviewed in detail to extract key information across several dimensions. These included the type of waste discussed, the specific technologies deployed, geographic and regulatory context, implementation outcomes, and

sustainability impact metrics or economic circularity. The extracted data were categorized into themes and mapped using an inductive approach to identify common patterns, technological trends, barriers, and opportunities. Technologies were analyzed not only by their function but also by the phase of the waste life-cycle in which they operate. These include sorting, collection, logistics, treatment, recycling, and final valorization.

The data analysis was structured around the three core technologies: AI, IoT, and Blockchain. AI applications, including both ML and DL techniques, were categorized based on their roles in waste prediction, classification, and optimization. IoT applications involved sensor-enabled bins, GPS tracking, and integration with data platforms for real-time WM. Blockchain use cases included waste traceability, decentralized waste trading, digital certifications, and smart contracts for compliance tracking.

2.4 | Review Credibility and Validation

The study adhered to the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) to ensure transparency and credibility in the review process, as reported by Araújo et al. (2020). The PRISMA framework, as presented in Figure 1, helped in documenting the flow of study selection, justification for exclusions, and consistency in the synthesis of findings. In addition to academic rigor, the review aligned its evaluation with globally recognized frameworks and policies. These included the United Nations SDGs, particularly Goals 7, 11, 12, and 13 (Alao et al. 2022; UN 2024). Relevant ISO standards, like ISO 14001 on environmental management (ISO 2015), and International Telecommunication Union (ITU) Smart City guidelines were also considered to assess the compliance and integration potential of reviewed technologies within policy and regulatory ecosystems (ITU 2021).

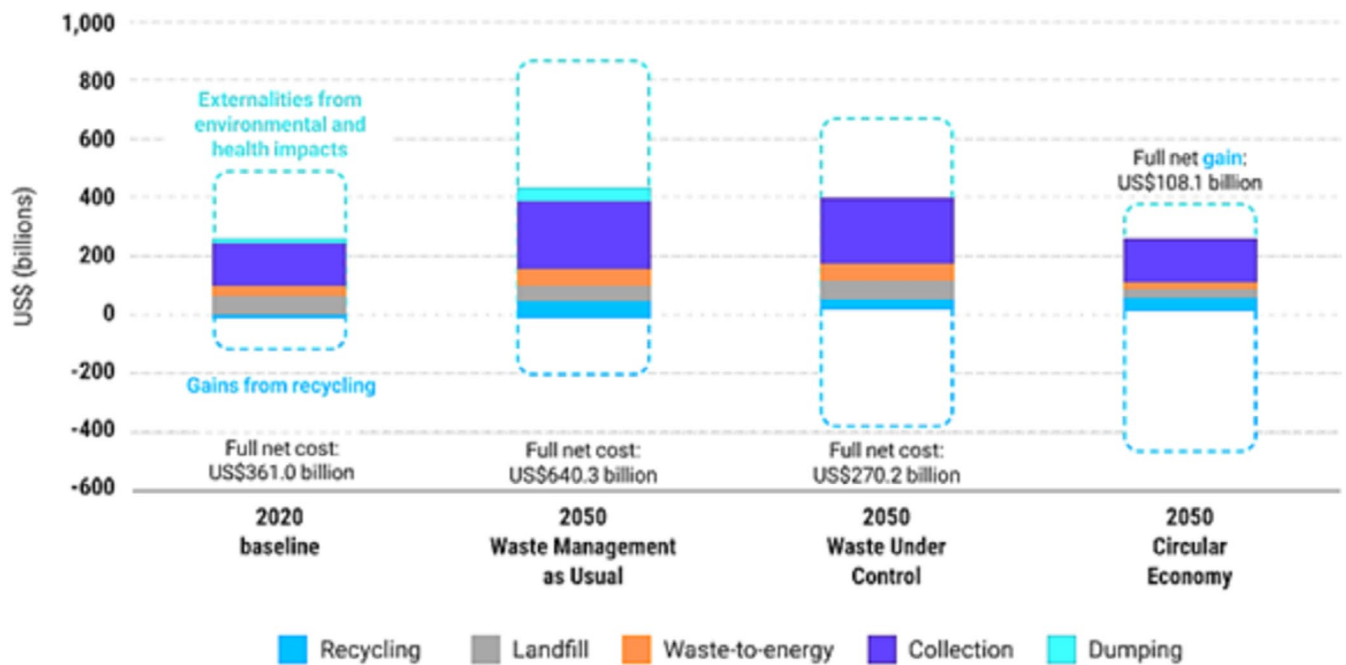


FIGURE 2 | Total cost of worldwide WM (UNEP 2024) (open access, no permission required).

3 | Overview of Global WM Challenges

The world generates an overwhelming volume of waste, and this is associated with the continuous rise in global population, increasing consumption, and urbanization. WM is among the critical challenges faced by different countries around the globe in recent times. The inability of the prevailing WM technology/systems to keep pace with this growth worsens the related environmental challenges, social disproportions, and economic inadequacies on a global scale.

The environmental concerns of improper WM are overwhelming. Landfills, which continue to be the main method of waste disposal in several countries, are the primary site for GHG emissions (Zhi et al. 2025). Unwanted organic materials decompose in the landfills and release GHG and methane. These emissions can potentially cause 25 times the global warming compared to carbon dioxide emissions over 100 years (Mundra and Lockley 2024). In addition, improper WM leads to air, soil, and water pollution. Plastics, for instance, cause environmental degradation in many countries, especially when they accumulate and form huge garbage in marine ecosystems, threaten biodiversity, and enter food chains (Fang et al. 2025). Also, e-waste contains harmful substances, including cadmium, mercury, and lead, which permeate the environment and cause significant damage to the health of the people (Jha et al. 2024).

From the socio-economic point of view, ineffectiveness is another major concern. Annually, valuable materials are discarded globally due to inadequate recovery and recycling methods. In some developing countries, a significant portion of the national budget is allocated to waste collection and disposal. However, these allocated funds fail to achieve the desired results due to technological inefficiency and poor infrastructure. Further, industries cannot often convert their waste into valuable products to save costs and for revenue. Discriminatory WM systems

unreasonably afflict marginalized populations. In many places around the world, people living close to landfills are exposed to harmful emissions, which pose severe health risks. Unskilled workers, who play a critical role in the collection, transportation, and recycling of waste in developing countries, most of the time work in unsafe conditions, with poor protective equipment (Rajendra 2024). This lack of regulation and unhealthy working conditions affects the poor and vulnerable in developing nations.

Figure 2 presents a comparative analysis of the economic implications of four global WM scenarios: the 2020 baseline, 2050 WM as usual, 2050 waste under control, and a 2050 CE model (UNEP 2024). The chart quantifies the full net costs or gains associated with each scenario by accounting for the expenses of core waste system components. These components include collection, recycling, landfill, waste-to-energy, and dumping. Also, the externalities from environmental and health impacts and the economic gains from recycling are included. In the 2020 baseline, the global waste system incurs a net cost of \$361 billion, driven primarily by high collection and waste-to-energy expenses, along with notable contributions from landfilling and dumping. While some economic gains are derived from recycling, they are not sufficient to offset the broader costs, especially when environmental externalities are included.

3.1 | Country and Sector-Specific Challenges

The summary of WM challenges across countries and sectors is presented in Table 1. In advanced countries, higher rates of waste recycling activities are offset by the popularity of non-decomposable waste, like electronic and plastic waste. While waste-to-energy technologies help to reduce dumping and open-field burning, they also raise environmental concerns due to GHG emissions and the disposal of leftover ash (Mubeen and Buekens 2025). On the contrary, developing countries face

TABLE 1 | Sector-specific WM challenges.

Sector	WM issues	Role of WV in addressing challenges	References
Agriculture	– Crop residue burning leads to air pollution and soil nutrient depletion.	– Conversion of crop residues into biochar, compost, or biogas to improve soil fertility and energy production.	Czekala et al. (2023); Ibitoye et al. (2021a); Ibitoye et al. (2021b)
	– Improper disposal of livestock manure causes water contamination and methane emissions.	– AD of manure to produce biogas and organic fertilizer.	
	– Inefficient utilization of agricultural waste results in resource wastage.	– Processing of waste into value-added products like biofertilizers, animal feed, or bioplastics.	
Manufacturing	– Hazardous waste (solvents, heavy metals) poses environmental risks.	– Adoption of cleaner production techniques and industrial symbiosis to repurpose byproducts.	Andre et al. (2021); Ibitoye et al. (2023); Shahabuddin et al. (2020)
	– Poor implementation of waste-to-energy technologies leads to waste accumulation.	– Gasification and pyrolysis to convert industrial waste into syngas, char, and bio-oil.	
	– Inefficient recycling increases material costs.	– Material recovery and reuse of production waste to reduce raw material dependence.	
Construction and Demolition	– High volumes of construction debris contribute to landfilling and illegal dumping.	– Recycling and repurposing of demolition waste into construction aggregates and eco-friendly bricks.	Alsheyab (2022); Buckner et al. (2016); Ramos and Martinho (2023)
	– Poor sorting and reuse of materials increase carbon footprints.	– Upcycling of waste materials (e.g., wood, metal) for new construction or interior design applications.	
	– Hazardous materials like asbestos and lead-based paints pose health risks.	– Safe disposal and alternative material development to replace hazardous substances.	
Healthcare	– Inadequate disposal of biomedical waste leads to disease outbreaks and water contamination.	– Implementation of safe waste sterilization and conversion into energy or recycled plastics.	Jayashree Dipak (2016); Keesagani et al. (2023)
	– Mixing of hazardous and general hospital waste complicates disposal.	– Development of specialized sorting and recycling systems for hospital waste streams.	
	– Lack of treatment facilities for medical plastics and expired drugs.	– Chemical recovery and transformation of pharmaceutical waste into reusable compounds.	
Electronics (E-Waste)	– Large volumes of obsolete electronics contain hazardous substances like lead and mercury.	– Urban mining to recover precious metals (gold, silver, rare earth elements) from discarded electronics.	Ankit et al. (2021); Bhutta et al. (2011); Jha et al. (2024)
	– Informal handling of e-waste leads to toxic exposure and groundwater contamination.	– Development of eco-friendly e-waste recycling facilities and safe extraction techniques.	
	– Low recovery rates for valuable metals result in resource depletion.	– Hydrometallurgical and pyrometallurgical methods for metal recovery.	

(Continues)

TABLE 1 | (Continued)

Sector	WM issues	Role of WV in addressing challenges	References
MSW	– Poor waste segregation contaminates recyclables and reduces recovery efficiency. – Heavy reliance on open dumping and landfilling leads to high methane emissions. – Weak policies and a lack of funding hinder modern WM solutions.	– Implementation of waste-to-energy plants and material recovery facilities. – Biogas recovery from landfills and conversion of organic waste into compost. – CE initiatives to incentivize recycling and reuse industries.	Alao et al. (2022); Ibitoye et al. (2021a); Kalyani and Pandey (2014); Traven (2023)
Textile and Fashion	– Fast fashion trends lead to massive landfill accumulation. – Synthetic fibers release microplastics into water bodies. – Blended fabrics are difficult to recycle.	– Upcycling and repurposing textile waste into new clothing or insulation materials. – Development of biodegradable fabrics and recycling programs for synthetic textiles. – Chemical and mechanical separation techniques for fabric recovery.	Bick et al. (2018); Teerawattananon et al. (2021)
Energy and Mining	– Large-scale mining waste causes land degradation and water contamination. – Disposal of coal ash poses risks of heavy metal leaching. – Inefficient resource recovery from drilling waste leads to pollution.	– Recovery of valuable minerals from tailings and extraction residues. – Conversion of coal ash into construction materials like cement or road fillers. – Reuse of drilling waste in industrial applications and energy recovery processes.	Bansah et al. (2024); Punia and Singh (2021); Rahman (2024)

different challenges of inadequate infrastructure for waste collection, separation, and conversion, which result in the dumping and burning of waste in open fields (Yan et al. 2020). The recent rapid population growth and urbanization in these regions worsen the problem. This is because the municipal WM cooperation struggles to handle the enormous waste generated due to rising populations and expanding cities.

3.2 | Addressing the Global Waste Crisis

Figure 3 (data extracted from the United Nations Environment Programme report [UNEP 2019]) provides a comprehensive overview of regional disparities in global WM strategies. It reports the performance, highlighting both the qualitative WM approaches and the quantitative distribution of treatment methods across major world regions. Figure 3 categorizes regions according to their dominant WM approach. At one end of the spectrum, West Asia and Africa are identified as regions that primarily rely on landfilling, with minimal implementation of waste recovery practices. These regions face structural and infrastructural limitations that hinder the adoption of sustainable waste solutions. Moving along the continuum, Latin America and the Caribbean are also landfill-reliant but show early signs of growing recovery initiatives, indicating a shift toward improved WM. North America represents a transitional region that

continues to rely significantly on landfilling but supports this with relatively well-established recovery programs, including recycling and incineration with energy recovery. Asia Pacific and the global average fall in the middle of the spectrum, with considerable use of landfills. However, the region displays increasing efforts to incorporate incineration and recycling into waste systems. At the most advanced end, Europe is characterized by its prioritization of recycling, composting, and energy recovery, with landfilling used only as a last resort. This positioning reflects Europe's regulatory frameworks, policy incentives, and infrastructure investments aligned with CE principles.

3.2.1 | The Need for Innovative Solutions in Waste Treatment

As the world struggles with the ever-increasing waste challenges, traditional approaches to WM, treatment, and conversion are proving insufficient to alleviate the environmental impact, complexity, and scale of contemporary waste streams. With global waste generation anticipated to increase to about 4 BT per year by 2050, it becomes very important to develop innovative solutions to address global waste generation (Kaza et al. 2018; Maalouf and Mavropoulos 2023). Technological advancements, combined with sustainability practices and resource efficiency, offer prospects to reconsider WM and transform waste

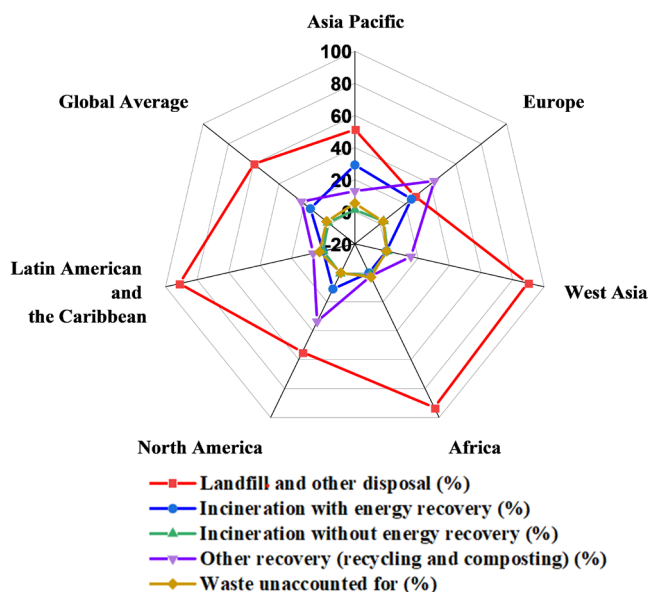


FIGURE 3 | Global WM methods by region.

challenges into opportunities for environmental transformation and revenue generation.

4 | Innovative Approaches to WM

The worldwide solid waste could grow by about 70% by 2050 (Negro et al. 2024). Such trends demand more sustainable and efficient WM strategies beyond conventional landfilling or incineration. As a result, researchers and industry professionals are investigating cutting-edge strategies that use innovative technologies to revolutionize waste handling and disposal. Digitalization has been embraced by recent WM initiatives, which have included blockchain technology, IoT, and AI technologies like ML and DL (Jiang et al. 2023). These new technologies have the potential to increase the sustainability, transparency, and efficiency of WM systems. They also improve resource recovery. The implementation of these technologies is accompanied by several limitations and difficulties. An outline of the main technological strategies advancing WM innovation is given in this section.

4.1 | Applications of DL in WV

DL, a subfield of ML that makes use of multifaceted neural networks, has developed into a powerful tool for automating tasks and recognizing patterns (Nguyen et al. 2021). WV is particularly pertinent to DL in the context of WM (Fang et al. 2023). A crucial step in many WV processes (such as recycling or composting) is the precise identification, sorting, and classification of waste materials. In the past, waste sorting has been laborious and prone to errors, which compromise recyclables and lead to insufficient resource recovery (Kannan 2025). DL provides a means to automate and significantly enhance this process, especially in the form of computer vision (Aral et al. 2018; Sayem et al. 2025). By training convolutional neural networks (CNNs) or other deep models on large datasets of waste images, systems can learn to classify different types of waste (plastics, paper,

glass, metal, organic, etc.) with high speed and reasonable accuracy (Malik et al. 2022; Sousa et al. 2019). DL models can also be applied to sensor data beyond images. For example, interpreting spectra from material sensors or learning patterns in waste generation data, but their most prominent use so far is in visual identification and robotics for waste sorting (Dokl et al. 2024).

4.1.1 | Technical Perspective on DL Models

Recent developments in computer vision have significantly advanced intelligent waste sorting, with modern object detection models playing a critical role in improving the speed and accuracy of classification processes. In particular, the YOLO (You Only Look Once) series has gained considerable attention for its real-time detection capabilities (Ahamed et al. 2025; Shad et al. 2025). Studies such as those by Riyadi et al. (2024) have evaluated both YOLOv5 and the more recent YOLOv8 and YOLOv9 in classifying recyclable materials. In addition to the YOLO models, other DL frameworks such as Residual Network with 50 layers (ResNet50) and Mask R-CNN have also been widely implemented in waste recognition tasks (Abdu et al. 2022; Sapkota et al. 2024). These models, known for their strong feature extraction capabilities and robustness in complex visual environments, further contribute to the growing ecosystem of AI-powered WM solutions. Each of these models exhibits unique architectural features, computational characteristics, and deployment profiles, making them suited to distinct stages or challenges in WV.

YOLOv5, created by Ultralytics, has become the industry standard for real-time object detection, especially in waste sorting applications where speed and precision are essential (Hussain 2024). Constructed using a Path Aggregation Network (PANet) neck and a Cross Stage Partial (CSP) backbone, the model provides a useful trade-off between computational efficiency and detection performance. Because of this, it is particularly useful in situations with limited processing power.

Recent studies underscore YOLOv5's effectiveness in handling trash classification tasks. For example, on a dataset containing 2500 images across five types of waste, the model achieved a precision of 90.6% and a recall of 83.3%, indicating its strong potential in identifying and categorizing waste materials accurately (Jalal et al. 2023). One of its strengths lies in its scalable design. The users can choose from different model sizes (small, medium, large, extra-large) based on specific performance or hardware requirements.

The efficiency of YOLOv5 in handling waste classification tasks is demonstrated by recent studies. Examples include the model that demonstrated a high capability for reliably detecting and classifying waste materials with a precision of 90.6% and a recall of 83.3% on a dataset of 2500 images of five different types of waste (Jalal et al. 2023). Its scalable design is one of its advantages- the users can select from small, medium, big, and extra-large model sizes according to hardware or performance requirements (Hussain 2024).

A practical illustration of this flexibility is the YOLOv5s variant, which reaches around 81% mean Average Precision

(mAP@50) on the widely used TACO dataset. With an inference time of just 17 milliseconds per image on an NVIDIA RTX GPU, it proves to be a strong candidate for integration into real-time, edge-based waste sorting systems (Hussain 2024). This kind of adaptability and efficiency makes YOLOv5 a valuable tool in efforts to streamline WM and improve recycling accuracy.

While YOLOv5 has already proven effective in detecting a variety of waste types (ranging from plastic and paper to glass, metal, cardboard, and organic materials), YOLOv8 has demonstrated even greater precision and faster processing times (Shad et al. 2025). These findings align with broader trends in smart WM, where advanced DL architectures are being adopted to enhance operational efficiency. Recent studies highlight how YOLOv5 can reliably identify common categories of recyclable and biodegradable waste. Meanwhile, Bawankule (2023) and Shad et al. (2025) reported that YOLOv8 outperforms several state-of-the-art detection algorithms, positioning it as a valuable asset in the automation of waste classification systems.

Unlike earlier YOLO versions, YOLOv8 adopts an anchor-free detection paradigm, simplifying the training process and improving object localization, particularly for small and irregularly shaped items (Yaseen 2024). It utilizes a CSPNet-based backbone for deeper feature extraction and combines Feature Pyramid Network (FPN) with Path Aggregation Network (PAN) for effective multi-scale detection (Yaseen 2024).

Performance evaluations on widely used benchmarks such as the COCO dataset and Roboflow 100 show that YOLOv8 delivers top-tier accuracy. Further, it maintains the real-time inference speeds needed for deployment on various hardware configurations (Yaseen 2024). It has been demonstrated that additional improvements like dynamic convolution layers and attention mechanisms greatly increase the capacity of the model to identify small objects. This is crucial for tasks like waste sorting, where visual cues may be poor or partially hidden.

In applied settings, YOLOv8 demonstrates flexibility through its various configurations. For instance, YOLOv8s performs well on smaller datasets with limited computational resources, making it suitable for edge devices in smart waste bins. Conversely, YOLOv8l offers higher mAP@50 values but requires more processing power, making it ideal for centralized sorting hubs (Shroff et al. 2023). These trade-offs allow users to tailor the model according to specific environmental and hardware constraints. Furthermore, YOLOv8's use of NMS-free training and spatial-channel decoupled down-sampling contributes to both computational efficiency and improved detection robustness, particularly in cluttered or complex visual environments (Hussain 2024). Taken together, these architectural improvements establish YOLOv8 as a state-of-the-art tool for real-time vision applications, especially in domains like intelligent waste sorting, where precision, speed, and adaptability are critical.

Recent research has highlighted the growing role of DL in advancing automated waste classification, with ResNet50 standing out as a particularly effective model. On the widely used

TrashNet dataset, ResNet50 achieved an accuracy of 84%, showcasing its strength in image-based classification tasks related to waste (Fatema and Maradithaya 2024). Performance improved further when ResNet50 was integrated with Inception modules, reaching an accuracy of 88.6% (Riyadi et al. 2024). These results reflect the advantage of combining different neural network architectures to enhance feature learning and classification precision.

An enhanced version of ResNet50 has also been shown to outperform the original model significantly. In a study by Ma et al. (2022), the improved version delivered a 7.62% gain in accuracy on the same dataset, underscoring the potential of architectural tuning in DL pipelines. Moreover, a comparative evaluation found that ResNet50 could reach up to 95% accuracy in certain waste classification tasks. The InceptionV3 achieved 100% accuracy across all tested categories, indicating its robustness in distinguishing visually similar waste items (Al-mashhadani 2023).

To mitigate the challenges posed by limited labeled data, common in real-world WM scenarios, researchers have also explored meta-learning approaches. For instance, the application of Model-Agnostic Meta-Learning (MAML) alongside ResNet50 has shown improved generalization capabilities. This results in minimal training iterations, making it a promising strategy for low-data environments (Fatema and Maradithaya 2024).

EfficientNet models have emerged as strong contenders in the field of automated waste classification. They offer a well-balanced trade-off between accuracy and computational efficiency. The architecture's core strength lies in its compound scaling technique, which systematically adjusts the model's depth, width, and input resolution to optimize performance across various resource constraints (Tan and Le 2019). This approach has since been refined to better capture inter-dimensional dependencies and adhere to real-time runtime requirements, further boosting both inference speed and classification accuracy (Lin et al. 2023).

In practical waste classification scenarios, EfficientNet-B0 has demonstrated promising results. Through transfer learning, the model achieved an accuracy of 91.94% on a six-category waste dataset. This underscores its capability to generalize effectively with limited training data (Ananda and Setyawan 2026). Building on this success, a specialized variant known as ScrapNet, based on the EfficientNet architecture, achieved remarkable performance metrics, scoring 98% accuracy on the TrashNet dataset and 92.87% on a more diverse, real-world dataset (Masand et al. 2021). What sets ScrapNet apart is its ability to deliver these results with a significantly smaller model footprint, highlighting EfficientNet's potential for deployment in resource-constrained environments.

Mask R-CNN stands out as a robust framework for object instance segmentation, building upon the capabilities of Faster R-CNN by introducing an additional branch to predict pixel-level masks for each detected object (He et al. 2020). This enhancement allows the model not only to identify and localize objects but also to segment them with high precision, making

it particularly valuable for complex image analysis tasks like waste detection.

Recent applications of Mask R-CNN in the domain of WM have demonstrated its versatility. In one study focusing on the classification of multiple household waste categories, the model achieved a mAP of 38.25%, a mAR of 65.50%, and an F1 score of 48.29%. These results show reasonable performance despite the inherent variability in waste images (Tomas et al. 2024). Beyond static classification, Mask R-CNN has also been deployed in more dynamic settings. For example, Aarth and Rishma (2023) used the model to detect and localize waste objects in natural outdoor environments, extracting key geometric features such as the object's centroid, orientation, and clamping points. These attributes are crucial for enabling robotic arms or automated systems to safely and accurately manipulate waste materials.

Details of these benchmarks, including the dataset used, inference time, and model size, are summarized in Table 2.

4.2 | Role of ML in Waste Transformation

ML refers to a broad class of algorithms that enable computers to learn from data and generate predictive or decision-making outputs (Gupta et al. 2024). Unlike DL (covered in Section 4.1), ML encompasses a wider set of approaches, such as random forests (RF), support vector machines (SVM), reinforcement learning (RL), and principal component analysis (PCA)-based hybrids (Figure 4a). These methods are increasingly applied to waste transformation processes, such as biological (e.g., composting, AD), thermal (e.g., incineration, pyrolysis, gasification), and chemical (e.g., chemical recycling of plastics). These processes are inherently complex and influenced by feedstock composition, temperature, catalysts, and other variables (Dodamepegama et al. 2024; Nath et al. 2025).

By learning from experimental or operational datasets, ML models can predict process yields (e.g., biogas production), identify optimal operating conditions, and minimize undesirable by-products (Gu et al. 2021). This data-driven approach complements traditional process engineering, which often relies on physics-based models and extensive laboratory testing. Recent work demonstrates the ability of ML models to forecast waste generation, optimize collection and routing, and support co-disposal strategies (Das et al. 2018; Kumar et al. 2024). The increasing availability of sensor data from modern waste facilities has further expanded the scope of ML applications, enabling real-time, adaptive, and more sustainable WM systems.

4.3 | IoT and WM

The IoT refers to networks of interconnected sensors and devices capable of collecting and transmitting data through local or wide-area connections. In WM, IoT underpins “smart WM” by embedding bins, collection vehicles, and treatment facilities with real-time monitoring capabilities (Malik 2023; Malik et al.

2022). This digital infrastructure addresses one of the sector's chronic challenges, the lack of timely and reliable information. For example, bin-level sensors can report fill status, enabling dynamic route optimization that reduces fuel consumption and prevents overflow (Ali et al. 2020; Ramson et al. 2021; Saha et al. 2017). Similarly, IoT devices monitor environmental conditions at landfills, track waste movements within material recovery facilities (MRFs), and provide auditable data streams for regulatory reporting.

As illustrated in Figure 4b, IoT acts as the “nervous system” of modern WM, creating transparency and linking multiple actors across the lifecycle, from households and municipalities to haulers, recyclers, and regulators. These data streams also serve as inputs to complementary technologies such as ML and blockchain, where analytics and traceability frameworks convert raw sensor signals into actionable insights. In this way, IoT not only enhances operational efficiency but also strengthens accountability and citizen trust in emerging CE systems.

4.4 | Enabling Transparency in WM Through Blockchain Technology

Blockchain technology is a form of distributed ledger technology (DLT) that allows data to be recorded in a secure, tamper-proof way across multiple nodes (participants). Each transaction or record is bundled into a “block” and linked to the previous block, forming a chain that is cryptographically secured. The key features of blockchain are decentralization, immutability, and transparency. In WM, these features directly address some longstanding challenges: lack of trust and transparency among stakeholders, difficulty tracing waste flows, and data fragmentation. Figure 5a highlights that WM involves numerous stakeholders. These include waste generators, collectors, transporters, recyclers, and regulators. Information about waste (such as quantities, transfers, and recycling rates) is often isolated within individual parties, and in some instances, deliberately concealed, especially in cases of illegal dumping or fraudulent recycling credit schemes. Blockchain offers a way to create a shared, trustworthy record of waste-related transactions and movements (Bułkowska et al. 2023; Alabdali 2025). By recording each step a waste item or batch takes, from generation to final disposal or recycling, on a blockchain, we can enable end-to-end traceability (Figure 5b). This means that a company can prove its waste was properly treated, or a recycler can assure that the materials they claim to recycle were recycled. Such transparency is increasingly important for regulatory compliance, environmental reporting, and fostering public trust (and investor confidence) in WM and recycling initiatives.

4.5 | Proposed Integrated Framework for Smart WV

Global WM systems are becoming more multifaceted and decentralized, necessitating integrated approaches that can scale, automate, and enforce compliance along the whole value chain. Although blockchain, IoT, and AI technologies

TABLE 2 | Comparative analysis of DL architectures in WV.

Model	Type	Dataset	Inference			Deployment suitability	Remark	References
			Accuracy/%mAP@50	Time (ms)	Params (M)			
YOLOv5s	Object Detection	TACO	81	17	7.2	Edge Devices	Fast inference, moderate accuracy	Hussain (2024); Jalal et al. (2023).
YOLOv8l	Object Detection	TrashNet	88.2	25	43.7	GPU Servers	High accuracy, slower inference	Shroff et al. (2023); Yaseen (2024)
Mask R-CNN	Instance Segmentation	Household Waste	38.25 65.50% (Recall)	120	44.0	GPU Only	High precision segmentation, slower	Aarthi and Rishma (2023); Tomas et al. (2024).
ResNet50 + MAML	Classification	TrashNet	95	35	23.5	GPU, Cloud	High accuracy with meta-learning	Fatema and Maradithaya (2024); Ma et al. (2022)
EfficientNet-B0	Classification	Custom 6-Class Waste	91.94%	22	5.3	Edge & Mobile	Excellent accuracy and compact size	Ananda and Setyawan (2026); Tan and Le (2019); Yu et al. (2024)
ScrapNet (EfficientNet-Based)	Classification	TrashNet & Diverse	98% (TrashNet), 92.87% (Diverse)	28	6.2	Flexible (Edge & Server)	Superior performance with fewer parameters	Masand et al. (2021)

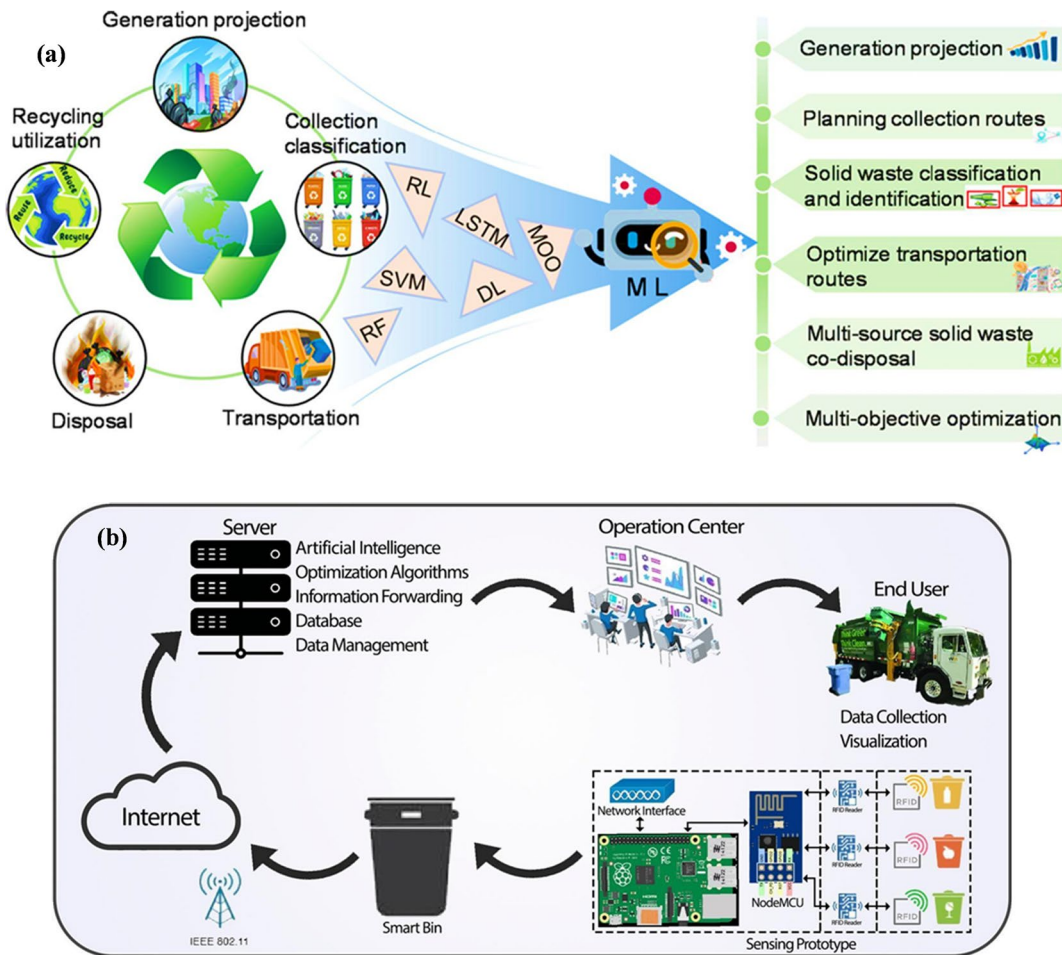


FIGURE 4 | (a) ML application in WM through prediction, classification, routing, and optimization (Zou et al. 2025) (with permission from Elsevier). (b) Schematic of IoT-Enabled Smart WM System (Mousavi et al. 2023) (with permission from Elsevier).

have all demonstrated promise in separate WM applications, this study suggests a novel unified architecture that combines these technologies into a coherent, interoperable framework. Through this integration, waste lifecycle management, which includes production, categorization, transportation, transformation, and regulatory compliance, becomes more traceable, intelligent, and trustworthy in addition to increasing operational efficiency.

The unified framework is structured into three interdependent layers (Sensing, Processing Layer, and Trust Layer), each contributing a distinct technological function to the overall WV system. The layers of the framework and their specific functions in the unified WV architecture are displayed in Table 3. The Sensing Layer forms the interface between the physical environment and the digital system, enabling automated data acquisition through embedded, networked devices. The Processing Layer uses AI-driven models for control, forecasting, and categorization to turn the unstructured information into actionable intelligence. The Trust Layer, which is at the top, uses automated enforcement procedures and decentralized documentation to guarantee transparency, verifiability, and compliance. These layers work together to establish a WM ecosystem that is efficient, smart, and transparent, supporting

stakeholder trust, optimization across systems, and real-time operations.

4.5.1 | Implementation Details for the Unified Architecture

The suggested implementation offers a reliable, interconnected, and transparent architecture by describing the communication frameworks, on-chain/off-chain partitioning, data models, failure mitigation, and governance.

Interfaces and communication: Low-power wide-area network connectivity, like LoRaWAN, is advised for distributed, low-bandwidth sensors at the sensing edge. MQTT is lightweight and capable of scaling for messaging from local aggregators to the cloud. 5G Ultra-Reliable Low-Latency Communication offers low-latency capabilities for high-bandwidth applications, like robotics or cameras at MRFs. Raw data should be kept off-chain, and only hashed digests of sizable sensor datasets should be secured on the blockchain trust layer to reduce costs.

Data models and serialization: GS1 Digital Link guarantees seamless interaction with product and package

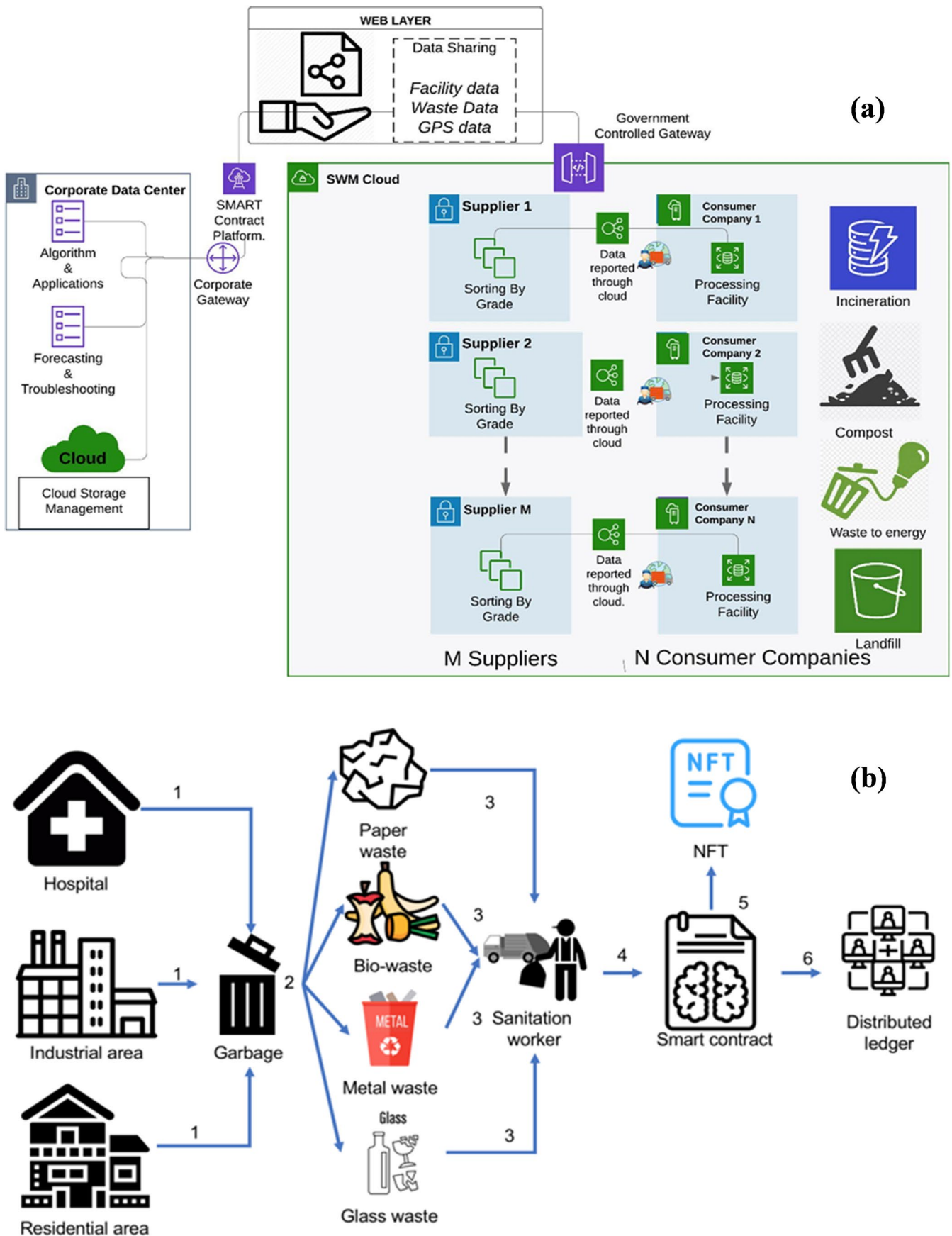


FIGURE 5 | (a) Illustration of blockchain-integrated smart WM ecosystem (Gopalakrishnan et al. 2021) (with permission from Elsevier). (b) Blockchain-Based Waste Traceability: From Source to Segregation via Smart Contracts and NFTs (Khiem et al. 2023) (open access, no permission required).

TABLE 3 | Framework layers and their functional roles in unified WV architecture.

Layer	Key technologies	Primary functions	Deployment context
Sensing Layer (IoT)	– Smart Bins – RFID Sensors – NIR Spectroscopy – GPS Trackers	– Real-time waste detection – Material identification – Asset tracking and geo-monitoring	Waste generation points, collection vehicles, sorting stations
Processing Layer (AI/ML)	– CNNs – Ensemble ML (RF, XGBoost) – Reinforcement Learning (RL)	– Waste classification – Predictive analytics (e.g., yield forecasting) – Dynamic decision-making	Edge devices (e.g., Jetson), cloud AI platforms, and control centers
Trust Layer (Blockchain)	– Smart Contracts – Distributed Ledger – Tokenized Incentives	– Immutable logging of transactions – Compliance automation – Reward-based user engagement	Cross-stakeholder networks, regulatory platforms, and digital wallets

identities throughout the supply chain, whereas JavaScript Object Notation for Linked Data (JSON-LD) should be the standard payload format for sensor metadata. To preserve semantic compatibility, waste categories should use the ontologies that are currently in use, such as EU waste codes or schema.org extensions.

Gas budget and on-chain/off-chain split: The only data kept on the chain are provenance records, cryptographic hashes, and smart contract events. High-volume sensor data should be stored in decentralized storage (like InterPlanetary File System) or the secure cloud. Batching, layer-2/consortium chain deployments, and event anchoring are suggested as cost-control measures.

Privacy, governance and roles: Role-driven access control ensures that municipalities, transport companies, recycling facility operators, recyclers, and citizens access only data and functions relevant to their responsibilities. Blockchain provides immutable, read-only records for transparency, regulatory compliance, and forensic auditing. Smart contracts enforce access rules, manage cryptographic keys, and automate permissions, ensuring secure data handling, clear accountability, and efficient governance.

Auditability and modes of failure: Failure modes that are expected include blockchain latency, DL incorrect classification, connectivity disruptions, and sensor drift or failure. Sanity checks, manual audit workflows for managing exceptions, periodic model retraining, redundant sensors, and backup rules for physical operations are some examples of mitigations.

4.5.2 | Workflow Illustration and Benefits

The unified framework uses a coordinated, automated series of actions to function. An integrated CNN classifier instantly recognizes recyclable waste when a household places it in a smart bin. Concurrently, embedded sensors measure the fill level of the bin and, when certain criteria are reached, initiate a collection

request via IoT gateway communication. Smart contracts are used to record the classification result and the pickup event on a blockchain ledger, guaranteeing tamper-proof tracking and traceability. In exchange for their compliant sorting behavior, the household is rewarded with digital tokens that are credited to their protected blockchain wallet. To promote ongoing advancement in system efficiency and responsiveness, aggregated data from bins, transport trucks, and treatment plants is then analyzed using ML algorithms to optimize collection routes, forecast future capacity requirements, and signal operational irregularities.

4.5.3 | Benefits of the Unified Architecture

Adaptive utilization of resources, predictive maintenance, and dynamic scheduling are made feasible using the unified architecture. In addition, WM becomes an intelligent, accountable, and participative process because of the numerous systemic benefits of the unified architecture. Every event, from trash generation to treatment, is verified and securely documented, ensuring end-to-end traceability. The system maintains compliance with local e-waste regulations and regulatory standards like ISO 14001. It drastically reduces manual intervention through the automation of classification, scheduling, and record-keeping (ISO 2015). Token-based incentives also encourage participation and engagement by gamifying appropriate disposal practices and encouraging wider public involvement. The platform may expand across a variety of contexts, from small cities to industrial hubs and even global waste exchange networks.

Figure 6 shows how the proposed unified architecture is organized as a multilayered data pipeline, with blockchain technology securing the outcomes, AI processes interpreting and acting upon the data, and physical sensors capturing real-world inputs. The system facilitates bidirectional flow, with data moving upward from the sensing layer to the trust layer and control signals, like the implementation of policies or corrective actions, flowing below when initiated automatically by

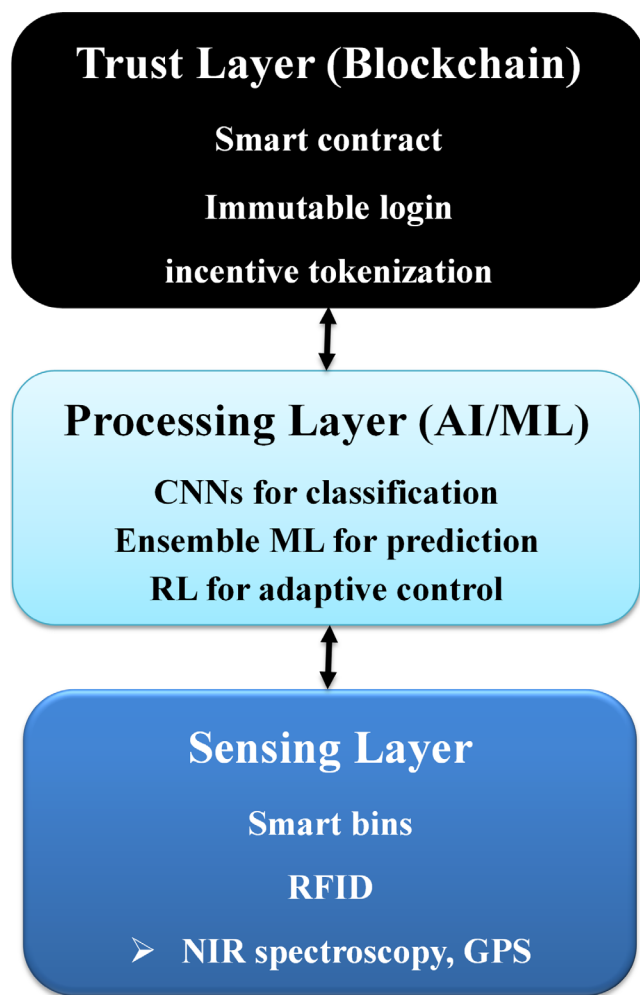


FIGURE 6 | Data flow and technological interaction across layers.

smart contracts. As a result, this unified architecture provides a cross-disciplinary, scalable design for integrating intelligent waste systems that are in line with the CE into the framework of future smart cities.

4.6 | Real-World Pilots, Outcomes, and Limitations

Although extensive literature exists on WV technologies, practical deployments remain comparatively limited. Recent pilot projects, however, provide concrete evidence of the benefits and challenges associated with integrating IoT, AI/ML/DL, and blockchain into WM systems.

4.6.1 | IoT-Enabled Waste Management Pilots

Investigations into IoT-enabled smart bins have reported substantial improvements in municipal waste collection efficiency across diverse urban environments. Mehta and Singh (2024) and Mehta and Rathour (2024) indicated that these systems can enhance collection efficiency by nearly 50% while reducing operational costs by 15%–25%. Sensor integration allows continuous monitoring of bin status, facilitating route optimization through

algorithmic approaches such as the Travelling Salesman Problem (Alnuaimi et al. 2024).

Beyond operational gains, environmental benefits have been documented. These include reductions in fuel consumption of up to 20% and CO₂ emissions ranging from 22% to 30% in cities such as Singapore, London, and New York (Mehta and Rathour 2024; Mehta and Singh 2024). Overflow incidents were reduced by 30%–83.3%, and recycling rates increased by 12%–25% (Mehta and Rathour 2024; Mehta and Singh 2024). Despite relatively high initial capital investment, a payback period of approximately 4 years supports the financial viability of smart bin adoption.

In Barcelona, IoT sensor networks integrated with cloud-based analytics have enabled real-time data acquisition and interpretation, contributing to waste reduction and improved citizen participation (Juneja 2025). Multi-year pilot programs confirmed the technical feasibility of cloud-integrated IoT analytics platforms, although challenges remain related to large data volumes and quality (Hasan et al. 2015).

4.6.2 | Blockchain-Enabled e-Waste and Traceability Systems

Blockchain-enabled systems integrated with IoT sensors have been shown to improve e-WM outcomes. About an 18% increase in material recovery efficiency and a 27% rise in recycling rates due to enhanced transparency and immutable audit trails have been reported (Ping et al. 2024). A blockchain-based e-waste recycling system implemented in Metropolis, a major urban center with a population of 5 million, demonstrated these benefits in practice (Ping et al. 2024). Early implementation faced stakeholder resistance stemming from data privacy concerns and operational disruptions. Structured engagement programs, including workshops and pilot demonstrations, gradually built trust and acceptance. Technical issues, such as sensor reliability under harsh facility conditions and blockchain scalability, were addressed via improved sensor design, redundancy, optimized consensus algorithms, and sharding techniques.

Limitations included the study's single-city scope over 12 months, potentially restricting generalizability to rural areas or developing regions and missing seasonal or long-term trends. Future research has emphasized exploring system scalability across diverse regulatory environments, cross-border e-WM, and the integration of AI for predictive analytics and process optimization. Data privacy remains a critical concern, with zero-knowledge proofs and homomorphic encryption proposed as enhancements (Ping et al. 2024).

Digital platforms leveraging blockchain for plastic traceability have also influenced consumer behavior, as access to recycling information and product provenance data motivates recycling decisions. In Italy, IoT and blockchain integration allowed producers to maintain control over electronic equipment until end-of-life while supporting CE objectives (Magrini et al. 2021). Reviews of blockchain in WM supply chains highlighted its

potential to strengthen traceability and accountability in closed-loop systems.

4.6.3 | AI- and IoT-Monitored Pyrolysis Systems

Significant advances have been observed in IoT-monitored and AI-assisted pyrolysis systems for biochar and bio-oil production. Cyber-physical integrations with pyrolysis reactors improved both yield and quality across various biomass feedstocks, with feedstock characteristics and reactor configurations being crucial determinants (Hersh et al. 2020). ML models enhanced predictive accuracy of biochar yield and composition by up to 12.7%, facilitating faster optimization (Li et al. 2022). Process parameters, including temperature, heating rate, reactor type, residence time, and feedstock blending ratios, were identified as critical factors affecting yield, stability, and carbon content (Al-Rumaihi et al. 2022). AI and process optimization improved efficiency and scalability in resource recovery systems (Thamarai et al. 2025).

Challenges remain due to feedstock heterogeneity, which varies by geographic origin, season, and post-harvest handling, complicating downstream process predictability (Thamarai et al. 2025). Limitations of AI include reliance on small or biased datasets, highlighting the need for standardized data protocols, open-access databases, transfer learning, and synthetic data augmentation (Thamarai et al. 2025). Scalability and technology transfer for emerging methods such as plasma-assisted conversion and microbial electrosynthesis (MES) are constrained by electrode degradation, mass transfer limitations, and reactor design complexity. Demonstrating MES integration into existing wastewater infrastructure has been proposed to improve commercial adoption. Regulatory fragmentation also limits the broader deployment of advanced biomass conversion technologies (Hersh et al. 2020).

4.6.4 | Limitations and Human Factors in the Technology Adoption

IoT and blockchain integration in WM face persistent barriers that reflect both technical and human factors observed in pilot programs. Technical limitations include connectivity disruptions, sensor calibration issues, and infrastructure constraints, particularly in densely populated urban areas (Gulyamov 2024; Sharma et al. 2020). Blockchain adoption is further constrained by high operational costs and complex governance requirements. This includes regulatory compliance and the need for standardization across stakeholders (Sharma et al. 2020; Torkayesh et al. 2022).

Human factors are equally critical. Inadequate training, low market acceptance, trust deficits, and reliance on manual processes often impede adoption. Staff may resist AI-driven recommendations despite demonstrated efficiency gains, and large-scale pilots frequently face challenges in implementing co-creation strategies due to technological immaturity and scale complexities (Gulyamov 2024; Padyab et al. 2020). Mitigation strategies include clear communication of technological

advantages, comprehensive capacity-building programs, and participatory governance to foster trust and facilitate deployment (Gulyamov 2024; Padyab et al. 2020; Sharma et al. 2020; Torkayesh et al. 2022).

Beyond technical and human limitations, funding mechanisms represent a critical barrier. High upfront capital requirements for IoT infrastructure, blockchain networks, and thermochemical facilities often limit adoption. Public-private partnerships (PPPs) and concessional finance schemes can mobilize resources, reduce perceived risk, and improve project bankability. Pilots show that targeted funding, especially when paired with government-backed incentives, increases adoption rates and encourages innovation (Sharma et al. 2020). Recommended mitigations include structured incentive programs, blended financing models, and transparent reporting to build investor confidence.

Regulatory hurdles also pose significant challenges. Data privacy concerns, cross-jurisdictional governance conflicts, and the legal ambiguity surrounding blockchain transactions impede deployment. For example, varying municipal regulations on e-waste tracking or smart bin data storage can hinder system interoperability. Mitigations include the establishment of regulatory sandboxes to test technologies under controlled conditions, harmonization of cross-jurisdictional policies, and development of blockchain legal frameworks to clarify liability and compliance responsibilities (Torkayesh et al. 2022).

Social impacts of technology adoption must not be overlooked. Automation in MRFs and the introduction of AI-based sorting systems may lead to job displacement among informal and formal waste workers. Failure to integrate these workers into new digital systems risks exacerbating social inequities. Recommended mitigations involve participatory planning, skill development programs, and inclusive business models that integrate informal sector actors into formalized workflows, ensuring economic and social benefits are distributed equitably (Gulyamov 2024; Padyab et al. 2020).

Although IoT, AI/ML, DL, blockchain, and thermochemical processes hold strong theoretical promise for WV, their practical value is best demonstrated in pilots and real-world trials. These projects show clear gains, lower costs, higher recovery, better accuracy, and improved transparency while exposing challenges such as high costs, governance gaps, data reliability, and user adoption. Table 4 presents a balanced overview of selected pilots across MSW, e-waste, medical waste, plastics, and biomass.

5 | Prospects

WM is undergoing a paradigm shift driven by the integration of advanced digital technologies. Traditional methods of managing, sorting, and transforming waste are increasingly being replaced or augmented by intelligent, automated systems. The urgency of addressing global waste challenges has led researchers and industry stakeholders to explore innovative approaches that combine DL, ML, IoT, and blockchain technologies. This

TABLE 4 | Selected pilots, outcomes, and limitations from literature.

Pilot/case	Waste type	Technology	Outcomes	Key limitations	References
Barcelona smart collection pilot	MSW	IoT fill-level sensors and GIS routing	- Reduced operational costs - Lower fuel use improved efficiency	- Data quality - Scaling challenges	Juneja (2025)
Indian smart wastes management pilot	MSW	IoT-enabled bins and GIS routing	Enhanced transparency	- High upfront cost - Unclear policy	Sharma et al. (2020)
Smart-bin pilots (Singapore, London, New York)	MSW	IoT smart bins and route optimization	- High collection efficiency ~50% - Reduction in costs 15%–25% - Less fuel utilization 20% - Reduction in CO₂ emission, 22%–30%	- High upfront cost ~4years payback - Connectivity issues	Mehra and Rathour (2024); Mehta and Singh (2024)
Blockchain–IoT for e-waste (Metropolis)	E-waste	IoT sensors and blockchain	- Higher Recovery up to 18% - High recycling rate, 27% - Better traceability	- Privacy concerns - Blockchain scalability - Limited to single city	Ping et al. (2024)
Italy EEE traceability	EEE	IoT and blockchain DLT	- Lifecycle control - Improved CE alignment	- Complex supply chains - Informal flows	Magrini et al. (2021)
Industrial pyrolysis–biochar demo (Europe/Asia)	Agricultural residues	IoT-monitored pyrolysis reactors	- Higher Biochar yield, 40%–50% - Improved bio-oil production, 20%–30% - Enhanced Syngas generation 25%–30% - Reduces CO₂ emissions	- Sensor drift at high temperature - Scaling barriers	Thamarai et al. (2025)
Pyrolysis pilots (biochar)	Biomass	IoT/AI-assisted pyrolysis	- High yield prediction accuracy 12.7% - Stable quality	- Feedstock heterogeneity - Small datasets	Hersh et al. (2020); Li et al. (2022)
AI/ML waste classification pilot	MSW sorting	DL models (CNN, hybrid)	94%–96% accuracy ~30% gain vs. manual sorting	Performance drops with soiled/heterogeneous inputs	Arun (2025).
Medical waste (Turkey)	Medical waste	IoT and blockchain	Identified adoption failures (training, transparency, staff acceptance)	Human/organizational barriers>technical	Torkayesh et al. (2022)
EU IoT LSPs	Cities, farming, wearables, transport	IoT pilots	Demonstrated economic/hedonic/social value	- Lack of Trust, Privacy issues - Immaturity - Co-creation difficult	Padyab et al. (2020)

Abbreviations: EEE—electrical and electronic equipment; GIS—geographic information system; LSP—large-scale pilots.

technological convergence represents the foundation of smart WM. By leveraging these tools, future systems could achieve high levels of automation, efficiency, and accountability in both urban and industrial settings. Moreover, such advancements are key enablers of the CE, facilitating material recovery, resource valorization, and sustainable consumption. However, realizing this potential requires addressing issues like model explainability, interoperability, energy efficiency, and regulatory compliance.

As DL systems become more prevalent, an important research and implementation consideration is how to better explain and trust AI decisions in WM. Efforts in AI explainability might produce tools that highlight which visual features led to a classification or that detect when the model is uncertain and flag those cases for human check—a hybrid approach that could maximize accuracy and safety.

Another direction is using advanced ML techniques like RL for process control (Croll et al. 2023). Instead of just predicting outcomes, an RL agent could learn control policies. According to Li et al. (2024), advanced control systems can reduce NOx emissions by up to 23.54% while maintaining optimal heat output through dynamic adjustment of operational parameters. Big data and real-time analytics will also influence ML in waste transformation.

Energy-efficient IoT is another focus, since WM involves deploying devices in the field, often battery-powered; there is ongoing work on extending battery life and using communication protocols that consume minimal energy. This will make long-term deployment easier and cheaper. The integration of IoT with AI is a clear direction.

Blockchain technology offers a novel means to inject transparency, security, and efficiency into WM processes. It aligns well with the goals of a CE by ensuring information about materials is transparently available throughout their lifecycle (Bułkowska et al. 2023). While challenges remain in scaling and integrating this technology, ongoing pilot projects and research indicate strong potential. If combined strategically with IoT data collection and AI-based analytics, blockchain could underpin a new era of accountable and transparent WM, giving all stakeholders—from citizens to regulators—greater confidence that waste is being handled in a sustainable manner that is claimed.

6 | Recommendation for Future Research

Several immediate and strategic actions are recommended to accelerate the transition towards intelligent, sustainable WM systems. Future research should focus on the synergistic integration of emerging technologies through real-world pilot implementations. While significant advancements have been made in isolated domains such as DL, IoT, blockchain, and ML, there remains a critical gap in unified, end-to-end frameworks that are validated under operational conditions. Researchers should prioritize the development of modular and interoperable architectures that can accommodate heterogeneous data sources, varied waste streams, and differing regulatory environments. Furthermore, the explainability and robustness of

AI models demand urgent attention, particularly in high-stakes applications such as hazardous waste classification or automated policy enforcement. Future research should investigate human-AI decision systems that guarantee extreme transparency and accuracy. The implementation of integrated digital work management platforms also requires the formalization of standards for data sharing, compatibility, and performance benchmarking.

Environmental impact and techno-economic analyses of smart waste technologies are also urgently needed. The lifecycle assessment, emissions, energy requirements, and societal implications of large-scale deployment of these technologies should all be assessed through thorough research. Solutions that can be adjusted to low-resource environments should be prioritized to make smart WM inclusive and available to everyone. Researchers should also investigate the socio-behavioral aspects of WM technology. Understanding user acceptance, ethical concerns, and governance frameworks will be vital for successful implementation. Cross-disciplinary collaborations, linking engineering, computer science, public policy, and behavioral sciences, are therefore strongly recommended to shape holistic, scalable, and sustainable solutions for the future of WV.

7 | Conclusion

Global MSW generation is projected to rise by nearly 70% and exceed 3.4 billion tonnes annually by 2050, making the transition from conventional disposal to technologically enabled WV both urgent and essential. This review provides an integrated framework spanning MSW, industrial residues, and e-waste, with emphasis on digital and thermochemical innovations.

Key findings demonstrate measurable efficiency gains. DL models achieved classification accuracies of up to 83.11% across 28 categories and mAP50 scores of 63% in complex detection scenarios. The WMR-DL framework improved material recovery by 30%, while IoT-enabled collection pilots reduced operational costs and fuel use by up to 20% (Arun 2025). Blockchain pilots increased traceability in plastics and e-waste recycling, and IoT-monitored pyrolysis reactors yielded 40%–50% biochar, 20%–30% bio-oil, and 25%–30% syngas.

Despite these advances, significant gaps persist: the lack of standardized datasets and benchmarks, limited deployment in resource-constrained settings, and the low interpretability of DL models. Additional barriers include regulatory complexity, interoperability challenges, and limited social acceptance.

Next steps are therefore prioritized as follows: (i) establishing shared datasets and benchmarking frameworks; (ii) validating pilot outcomes across diverse regions and waste streams; and (iii) developing policy and funding roadmaps to connect technological innovation with large-scale implementation.

This review combines quantitative evidence with a forward-looking agenda. It positions digital and thermochemical WV as a viable pathway for multiple benefits. These include advancing CE goals, reducing greenhouse gas emissions, and enhancing environmental resilience.

Technology	Description/summary
Deep Learning	Automated waste image classification and robotic sorting; anomaly detection in heterogeneous waste streams
Machine Learning	Predictive modeling for waste generation, routing optimization, and yield prediction in thermal/biological processes
Internet of Things	Fill-level smart bins, GPS-enabled fleets, and real-time process monitoring of reactors
Blockchain	Waste traceability, recycling credits, provenance tracking, governance
Pyrolysis	Thermochemical conversion of biomass and MSW into char, oil, and syngas
Gasification	High-temperature conversion of MSW/biomass into syngas for energy
Anaerobic Digestion	Biological decomposition of organic waste to produce biogas and digestate
Hydrothermal Liquefaction	Conversion of wet biomass into bio-crude under high temperature/pressure in aqueous phase

Author Contributions

Segun E. Ibitoye conceptualized and designed the review. The review was conducted by Segun E. Ibitoye and Amit K. Ball. Formal analysis was performed by Segun E. Ibitoye, Amit K. Ball, and Raul V. Rodriguez. Segun E. Ibitoye and Amit K. Ball prepared the original draft. Review and editing were carried out by Esther T. Akinlabi, Olalekan A. Olayemi, Peter O. Omoniyi, Pepluis Esteva, Raul V. Rodriguez, and Isaac K. Adegun. All authors have read and approved the final manuscript.

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